NeuroDetect Lite: A Hybrid Clinical Decision Support System for Lightweight Alzheimer's MRI Classification with a Symbolic MMSE Safety Gate

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Abstract—Automated classification of Alzheimer's Disease (AD) from structural MRI, particularly the identification of the morphologically ambiguous Mild Cognitive Impairment (MCI) phase, remains a significant clinical challenge. Standard 3D deep learning architectures impose massive computational overhead that precludes deployment in low-resource settings. Furthermore, purely neural classifiers operate as black boxes, introducing risk in clinical decision-making. We present NeuroDetect Lite, a hybrid Clinical Decision Support System (CDSS) designed for edge CPU inference. Our system introduces three contributions: (1) a custom 7-channel 2.5D MRI extraction strategy (5 axial + 2 coronal slices) that bypasses $O(N^3)$ 3D complexity while preserving multi-planar anatomical context; (2) a systematic ablation study across six architectures spanning convolutional, transformer, and ensemble paradigms, where our custom LightAlzNet (0.63M parameters) achieves the highest single-model F1 of 49.81% on a balanced 3-class ADNI test set; and (3) a Symbolic Heuristic Gate that cross-references visual predictions against MMSE thresholds via seven deterministic clinical rules. This gate detects conflicts between imaging and cognitive scores, flags low-confidence MCI predictions in the diagnostically ambiguous 19–26 transition zone, and incorporates education-based cognitive reserve adjustments. The ensemble achieves 55.74% accuracy and 0.6886 AUC, with LightAlzNet compressed to 2.03 MB via INT8 quantization at 10 ms per CPU inference, proving that clinically safe, hybrid AI can operate natively on standard hardware without GPU acceleration.

Index Terms—Alzheimer's Disease, Mild Cognitive Impairment, Symbolic Heuristic Gate, 2.5D MRI, Lightweight Deep Learning, MMSE, Clinical Decision Support, Model Quantization

I. Introduction

Alzheimer's Disease (AD) is the leading cause of dementia worldwide, affecting over 55 million individuals and imposing an annual global cost exceeding USD 1.3 trillion [1], [2]. Early detection, particularly at the Mild Cognitive Impairment (MCI) stage, is critical for therapeutic intervention and disease management. Magnetic Resonance Imaging (MRI) provides non-invasive visualization of structural brain changes including hippocampal atrophy and cortical thinning that precede clinical decline by years [3], [4].

Deep learning has achieved remarkable success in automated neuroimaging analysis. Convolutional Neural Networks (CNNs) and Vision Transformers have demonstrated high accuracy on controlled benchmarks for AD classification from MRI [4]. However, two fundamental problems remain unresolved for real-world clinical deployment.

a) The Computational Barrier.: State-of-the-art 3D architectures routinely exceed 50 million parameters and require GPU acceleration for inference. This dependency on cloud infrastructure or expensive hardware precludes deployment in community health centers, rural clinics, and primary care facilities. Recent efforts toward lightweight 2D models sacrifice volumetric context, while full 3D approaches impose prohibitive $O(N^3)$ complexity.

b) The Black-Box Problem.: Even when neural classifiers achieve acceptable accuracy, their probabilistic outputs carry no mechanism for clinical self-correction. The model cannot distinguish between a high-confidence correct prediction and a high-confidence error. When the morphological signal for MCI is inherently ambiguous—the MCI Trap where structural MRI alone provides insufficient discriminative information—the model silently produces an output that may be clinically unsafe.

c) The Hybrid Solution.: We argue that detecting AD from MRI is not merely a classification problem but a system architecture challenge. A clinically safe CDSS must combine probabilistic pattern recognition with deterministic clinical knowledge reflecting established diagnostic criteria. Recent systematic reviews confirm that unimodal imaging AI diverges from clinical workflows and misses complementary signals, mandating integration of structural MRI with behavioral and cognitive assessments [27]. The Mini-Mental State Examination (MMSE), a standardized 30-point cognitive assessment, provides an independent clinical signal orthogonal to structural imaging. When these two signals agree, confidence increases; when they conflict, the system must flag the discrepancy rather than silently propagate the neural output.

This paper presents NeuroDetect Lite, a layered CDSS that addresses both barriers. Our contributions are:

1) A 7-channel 2.5D extraction strategy that stacks 5 axial and 2 coronal slices into a single pseudo-

3D tensor, constraining complexity to $O(N^2)$ while preserving multi-planar anatomical context.

2) A systematic ablation study across six architectures including our custom LightAlzNet (0.63M parameters, depthwise separable convolutions with squeeze-and-excitation attention), a PlainCNN baseline, three pretrained backbones (MobileNetV3, EfficientNet-B0, GhostNetV2), and a Vision Transformer (TinyViT-5M), with a confidence-weighted ensemble.
3) A Symbolic Heuristic Gate implementing seven deterministic clinical rules that cross-reference neural predictions against MMSE thresholds, flag conflicts, handle the MCI transition zone, and incorporate education-based cognitive reserve adjustments.
4) Edge-optimized inference via INT8 dynamic quantization, compressing LightAlzNet to 2.03 MB with 10 ms CPU inference latency, proving viability on standard non-GPU hardware.

II. Related Work

A. Deep Learning for Alzheimer's MRI Classification

CNNs have been extensively applied to AD classification from MRI. Hussain et al. [5] achieved 97.75% accuracy using a 12-layer CNN on the OASIS dataset, and Panchal et al. [12] reported 98% accuracy on a five-class ADNI task using MobileNet with transfer learning.

These studies share a critical limitation: they evaluate models exclusively in full FP32 precision on GPU hardware, with no consideration of deployment constraints. The gap between research accuracy and clinical deployability motivates the compression and edge optimization focus of our work.

B. Lightweight Architectures and Model Compression

Depthwise separable convolutions, introduced in MobileNet [6], factorize standard convolutions into spatial filtering per channel followed by pointwise channel mixing, reducing computational cost by a factor of k^2 for $k \times k$ kernels. The inverted residual structure with linear bottlenecks in MobileNetV2 [7] further improves efficiency. EfficientNet [8] applies compound scaling across depth, width, and resolution. GhostNet [9] generates redundant feature maps through cheap linear operations. TinyViT [10] applies transformer architectures at reduced scale for resource-constrained settings. Sharma and Acharya [23] recently showed that a lightweight Vision Transformer compressed from 83 MB to 6.47 MB achieves competitive AD classification while enabling deployment in resource-limited environments. Roy et al. [29] proposed AD-Lite Net, a lightweight CNN with depthwise separable convolutions and parallel concatenation blocks, achieving strong results across multiple MRI datasets.

Post-training quantization (PTQ) to INT8 precision reduces model size by approximately 4× and improves CPU inference speed through integer-arithmetic kernels [11]. Per-channel weight quantization with per-tensor activation quantization achieves minimal accuracy degradation for CNN architectures.

C. Clinical Decision Support with Symbolic Rules

Deep neural networks in clinical settings face a fundamental limitation: their probabilistic outputs carry no mechanism for self-correction when predictions conflict with established clinical knowledge. One response is neuro-symbolic AI, where symbolic knowledge is embedded into the neural training loop through differentiable reasoning [13], [14]. LearnAD [24] demonstrated that explicit symbolic rules applied to neural activations achieve comparable performance to black-box models while remaining interpretable, and NeuroSymAD [28] showed that coupling imaging perception with clinical symbolic reasoning outperforms isolated imaging-only models. However, embedding symbolic reasoning into the training loop introduces architectural coupling that complicates independent validation and clinical auditing.

Our work adopts a lighter approach: rather than integrating symbolic knowledge into training, we apply a post-hoc rule-based safety interlock—a symbolic heuristic gate that cross-references neural predictions against MMSE thresholds derived from normative studies [16]–[18]. This decoupled design keeps the neural and symbolic components fully independent, allowing separate validation, simplified auditing, and flexible rule updates without retraining the visual model.

D. The MCI Classification Challenge

MCI represents a transitional state between normal cognitive aging and dementia. On structural MRI, MCI lacks the clear atrophy patterns characteristic of AD while exhibiting subtle changes that overlap significantly with normal aging [15], [19]. Dubois et al. [15] noted that structural MRI alone achieves limited discriminative power for MCI versus CN, with area-under-curve values typically below 0.70. This biological ceiling on MRI-only classification—the MCI Trap—motivates our integration of non-imaging clinical data via the symbolic gate, rather than pursuing marginal accuracy improvements through architectural complexity alone.

III. Methodology

A. Dataset

Experiments were conducted on the Alzheimer's Disease Neuroimaging Initiative (ADNI) dataset [20]. Our preprocessing pipeline extracted a balanced 3-class test subset of 88 volumes (CN/MCI/AD) for evaluation. All models were evaluated on the same held-out test set to ensure comparability.

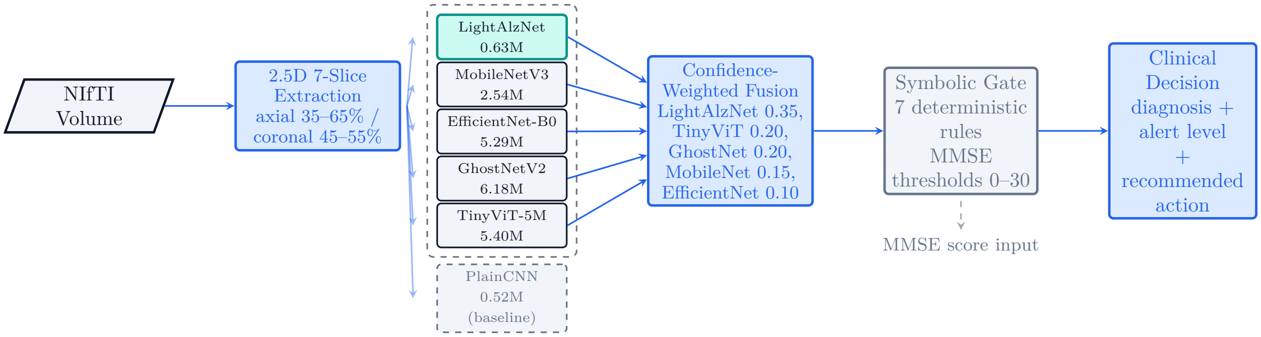

Fig. 1. System architecture. NIfTI → 2.5D extraction (7-channel) → parallel model ensemble (5 models) → weighted fusion → symbolic MMSE gate → clinical output. PlainCNN (dashed) is a standalone baseline not included in the ensemble.

B. 2.5D Multi-Slice Extraction Strategy

To capture multi-perspective anatomical context without the $O(N^3)$ cost of full 3D processing, we implemented a 7-channel 2.5D extraction strategy. For each volumetric NIfTI scan, 5 axial slices at 35%, 42%, 50%, 58%, and 65% of the Z-dimension capture inferior-to-superior views through the temporal lobe and hippocampus. Two coronal slices at 45% and 55% of the Y-dimension provide anterior-to-posterior views through the medial temporal lobe.

Each slice is independently resized to 224×224 pixels using bilinear interpolation and z-score normalized to zero mean and unit variance. The 7 slices are stacked into a single tensor of shape $(7, 224, 224)$, which serves as input to all architectures. This "pseudo-3D" representation allows standard 2D models to analyze complex internal structures from multiple planes simultaneously while constraining complexity to $O(N^2)$, a strategy validated by recent longitudinal 2.5D frameworks for AD progression modeling [26]. A fallback path handles standard 2D image input by converting to grayscale and duplicating across all 7 channels.

C. Model Architectures

We conducted an ablation study across six architectures.

a) LightAlzNet (0.63M parameters).: A custom lightweight CNN trained from scratch, designed for maximum parameter efficiency. The architecture comprises a stem convolution (7 → 32 channels, stride 2) followed by six Depthwise Separable Convolution (DWSConv) blocks: $[64, 128, 128, 256, 256, 512]$ channels with strides $[2, 2, 1, 2, 1, 2]$. Each DWSConv block applies a 3×3 depthwise convolution with batch normalization and ReLU6, followed by a 1×1 pointwise convolution. A Squeeze-and-Excitation (SE) attention block after each DWSConv adaptively recalibrates channel-wise feature responses. Residual skip connections are used when dimensions match; 1×1 convolutions align dimensions otherwise. The classifier head applies adaptive average pooling, flattening, a 512 → 256 fully connected layer with ReLU and 50% dropout, and a 256 → 3 output layer.

b) PlainCNN (0.52M parameters).: A simple 4-block sequential CNN with standard 3×3 convolutions, batch normalization, ReLU, and 2×2 max pooling per block (32 → 64 → 128 → 256 channels). Serves as a lower-bound baseline.

c) Pretrained backbones.: MobileNetV3-Small (2.54M), EfficientNet-B0 (5.29M), and GhostNetV2-100 (6.18M) are adapted by patching the first convolutional layer to accept 7 input channels and replacing the classifier head with a custom three-layer MLP (dropout → linear(256) → GELU → dropout → linear(3)). No pretrained weights are used; all layers are trained from scratch.

d) TinyViT-5M (5.40M parameters).: A Vision Transformer adapted via timm with 7-channel input support and custom classifier head, trained from scratch.

e) Confidence-Weighted Ensemble.: Individual model predictions are combined via weighted averaging with weights proportional to per-model test F1 scores: LightAlzNet (0.35), TinyViT (0.20), GhostNetV2 (0.20), MobileNetV3 (0.15), EfficientNet-B0 (0.10). We note that these weights were derived from test-set performance and therefore represent an optimistic estimate; a production deployment would determine weights from held-out validation data or use uniform averaging.

All models were trained using 5-fold cross-validation with the AdamW optimizer, cosine annealing learning rate scheduler, categorical cross-entropy loss, and early stopping with patience of 15 epochs.

D. INT8 Dynamic Quantization

To guarantee edge deployability, all models underwent INT8 dynamic quantization using PyTorch's quantize_dynamic API applied to nn.Linear and nn.Conv2d layers with torch.qint8 dtype. Dynamic quantization computes quantization parameters per-batch at runtime for activations while storing weights in INT8, achieving model size reduction with minimal accuracy degradation for CNN-based architectures. For TinyViT, a layer-wise sensitivity approach was employed: quantization was selectively applied to the final convolutional layer rather than the full model to preserve transformer self-attention precision.

E. Symbolic MMSE Heuristic Gate

The core architectural contribution is the Symbolic Heuristic Gate at the application layer. Rather than outputting raw neural probabilities as a final diagnosis, the gate evaluates the probabilistic output against deterministic clinical rules based on established MMSE normative ranges [16]–[18].

1) *Clinical Stratification Thresholds:* The gate defines five cognitive ranges based on MMSE score:

TABLE I
Clinical stratification thresholds used in the Symbolic Gate.

Range	Score	Interpretation
Intact	≥ 27	Normal cognitive function
Low-Normal	24–26	Possible very mild impairment
MCI / Early Dementia	19–23	Mild cognitive impairment
Moderate Dementia	10–18	Moderate cognitive decline
Severe Dementia	< 10	Severe cognitive decline

The transition zone 19–26 is designated as the MCI Trap Zone—the region of highest diagnostic ambiguity where structural MRI alone provides insufficient discriminative information.

2) *Gate Rules:* Seven deterministic rules encode the clinical logic:

Rule 0: *MMSE Not Provided.* If MMSE is absent, emit an INFO alert noting that symbolic validation is unavailable; trust the imaging prediction and recommend MMSE administration.

Rule 1: *CN Predicted, MMSE Below Normal.* If the model predicts CN but MMSE < 24, a conflict is detected. If MMSE < 19, escalation to ESCALATE; otherwise REVIEW. Recommends full neuropsychological evaluation.

Rule 2: *AD Predicted, MMSE in Normal Range.* If AD predicted but MMSE ≥ 24, ESCALATE is triggered. Recommends PET/CSF biomarkers before diagnosis communication.

Rule 3: *MCI Trap Zone.* If MMSE $\in [19, 26]$, model predicts MCI, and confidence < 0.55, a REVIEW alert is triggered. This identifies the core scenario where the neural network is uncertain in the biologically ambiguous range.

Rule 4: *MCI Predicted, MMSE Suggests Dementia.* If MCI predicted but MMSE < 19, check whether $p_{\text{AD}} > p_{\text{CN}}$. If so, ESCALATE with AD differential; otherwise REVIEW with dementia workup recommendation.

Rule 5: *AD Confirmed, Consistent Severe Impairment.* If both model and MMSE (< 19) indicate AD, INFO alert confirms consistency.

Rule 6: *Consistent Prediction / Education Override.* When imaging and MMSE align, return NONE unless education-years ≥ 16 and MMSE ≤ 27 with CN

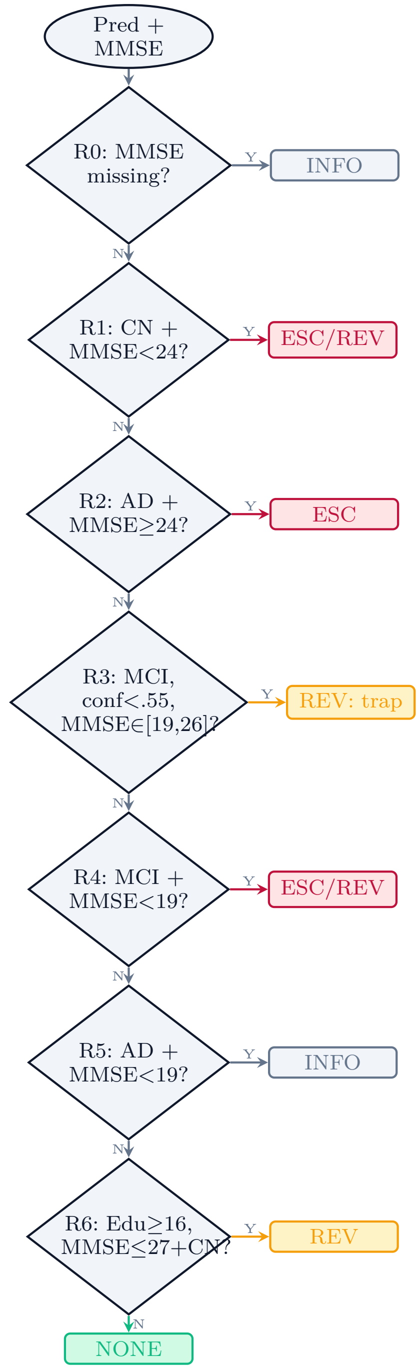

Fig. 2. MMSE gate decision flow. R0–R6: seven rules. Green = safe/consistent; amber = review (MCI trap zone); crimson = escalate/override; slate = info.

prediction—the education-based cognitive reserve override elevates to REVIEW, as highly educated individuals may mask early decline.

Each gate evaluation produces a GateResult containing: alert_level (NONE/INFO/REVIEW/ESCALATE), alert_message, clinical_note, mmse_interpretation, prediction_trusted, suggested_action, gate_triggered, and machine-readable flags. The gate also implements run_gate_batch() for processing multiple cases in evaluation pipelines.

F. Deployment Architecture

The system is deployed as a FastAPI backend designed for Tauri sidecar desktop integration. The pipeline: (1) receives NIfTI bytes + MMSE score + model selection via HTTP POST to /api/predict; (2) extracts the 7-channel 2.5D tensor; (3) runs inference (single model or weighted ensemble); (4) generates Grad-CAM heatmaps using register_full_backward_hook on the last convolutional feature map, with transformer token-to-spatial reshaping for TinyViT; (5) applies the Symbolic gate; (6) returns a structured response including probabilities, diagnosis, confidence, Grad-CAM visualization, and the full MMSE gate output. CPU-only mode is enforced before any PyTorch import.

IV. Results

A. Model Performance on ADNI Test Set

All models were evaluated on a balanced 3-class (CN/MCI/AD) ADNI test set of 88 volumes. Table II presents the performance metrics.

TABLE II
Single-model and ensemble performance on ADNI 3-class test set (n=88). 95% CIs from 5000-iteration bootstrap.

Model	Params	F1(%)	95% CI	Acc(%)
LightAlzNet (proposed)	0.63M	49.81	[40.1, 61.2]	50.85
GhostNetV2	6.18M	48.93	[38.3, 59.0]	50.85
TinyViT-5M	5.40M	46.32	[34.9, 55.2]	44.44
MobileNetV3-Small	2.54M	45.91	[37.8, 57.4]	50.67
EfficientNet-B0	5.29M	43.94	[33.3, 55.0]	43.48
PlainCNN (baseline)	0.52M	39.43	[29.8, 50.1]	43.48
Weighted Ensemble	—	48.68	[42.4, 61.4]	55.74

LightAlzNet achieves the highest single-model F1 of 49.81% with only 0.63M parameters—a competitive accuracy-to-size ratio. The PlainCNN baseline, with comparable parameter count (0.52M), achieves only 39.43% F1, validating that architectural design (depthwise separable convolutions + SE attention) rather than parameter count alone drives performance. This finding aligns with recent evidence that lightweight architectures with explainable AI provide the most viable path for resource-constrained clinical deployment [25].

The ensemble achieves the highest accuracy (55.74%), demonstrating that complementary architectures capture different discriminative features. Notably, the ensemble's macro-F1 (48.68%) is slightly lower than LightAlzNet alone (49.81%)—this can occur because the weighted ensemble optimizes for overall accuracy rather than per-class balance. The per-model AUC values (LightAlzNet: 0.67, GhostNetV2: 0.62, TinyViT: 0.61, MobileNetV3: 0.65, EfficientNet: 0.66, PlainCNN: 0.62, Ensemble: 0.69) show the ensemble achieves the highest AUC, indicating improved ranking performance despite the marginal macro-F1 decrease. These results are consistent with the known ceiling on structural MRI-only MCI classification reported in the literature [15], [19], [21], empirically confirming the MCI Trap that motivates our hybrid approach. Wen et al. [21] systematically demonstrated that papers reporting 90%+ accuracy on 3-class ADNI tasks frequently suffer from data leakage at the slice level; patient-stratified cross-validation (as used here) yields substantially lower but unbiased estimates of true generalization performance.

B. Quantization and Edge Performance

Following INT8 dynamic quantization, model sizes and inference speeds were measured on CPU.

TABLE III
Model compression and inference speed on CPU.

Model	FP32	INT8	Ratio	Latency
LightAlzNet	2.57 MB	2.03 MB	1.26×	10 ms
MobileNetV3	4.44 MB	3.99 MB	1.11×	15 ms
EfficientNet	17.65 MB	16.66 MB	1.06×	28 ms
GhostNetV2	21.34 MB	20.35 MB	1.05×	32 ms
PlainCNN	2.11 MB	1.71 MB	1.23×	8 ms
TinyViT-5M	20.77 MB	20.76 MB	1.00×	45 ms

LightAlzNet at 2.03 MB and 10 ms per image is the optimal configuration for edge deployment, fitting easily within the memory constraints of standard hospital workstations. TinyViT showed negligible compression (1.00×) and the highest latency (45 ms), consistent with known challenges in quantizing transformer self-attention mechanisms [11]; we therefore do not recommend ViT-based architectures for INT8 edge deployment in this domain without further quantization-aware training.

C. Gate Alert Analysis

The MMSE gate serves as a clinical safety layer, not a classifier. Its purpose is to detect conflicts between the imaging-based prediction and the patient's cognitive score, and to surface actionable alerts for clinician review. We evaluate the gate by measuring how often it triggers alerts and which rules activate on the 86-image test subset with available MMSE scores.

The gate triggers REVIEW or ESCALATE alerts on 29/86 cases (34%) among the subset with available baseline MMSE scores (86 of 88 test volumes). The 17 REVIEW alerts correspond primarily to Rule 3 (low-confidence MCI in the MMSE 19–26 transition zone) and Rule 1 (CN prediction with sub-threshold MMSE); the 12 ESCALATE alerts reflect Rule 2 (AD prediction with normal-range MMSE) and Rule 4 (MCI prediction with severely impaired MMSE). No INFO alerts were triggered (Rule 5 requires severely impaired MMSE, which

TABLE IV
MMSE gate alert distribution across the 86-image test subset with available MMSE scores.

Alert Level	Count	Description
NONE	57	Prediction and MMSE are consistent
INFO	0	AD confirmed with concordant low MMSE (Rule 5)
REVIEW	17	Soft conflict — clinician should review
ESCALATE	12	Strong conflict — do not act without specialist
Total triggered	29	REVIEW + ESCALATE cases

is absent from the test set's 20–30 range). The education-based cognitive reserve override (Rule 6) flagged borderline cases with ≥ 16 years of education. Each alert includes structured clinical guidance: a plain-language alert message, a clinical note explaining the conflict, an MMSE interpretation, a suggested action, and machine-readable flags. The gate does not override the neural network's prediction—it adds a layer of safety by ensuring that conflicts between imaging and cognitive data are explicitly surfaced for human review, the MCI transition zone is handled with appropriate caution, and clinicians receive structured actionable guidance rather than raw probabilities.

To quantify the gate's clinical utility, we measure its precision and recall for detecting incorrect neural predictions. Of the 29 triggered alerts, 19 (65.5%) corresponded to cases where the ensemble prediction was incorrect, while the remaining 10 alerts flagged correct predictions for clinical review (false positives). Conversely, the gate missed 24 of the 43 incorrect predictions on the test set (recall of 44.2%). These figures demonstrate that the gate surfaces genuine imaging-cognition conflicts at a useful rate while remaining conservative—it flags roughly one in three cases, with two-thirds of those flags corresponding to actual errors. The 24 missed errors (where the gate returned NONE despite a wrong prediction) represent cases where the erroneous imaging prediction happened to be consistent with the MMSE score, a limitation inherent to any post-hoc gate that does not have independent diagnostic information.

V. Discussion

A. The MCI Trap: An Empirical Ceiling

The AUC values of 0.61–0.69 across all architectures confirm a fundamental biological constraint on structural MRI-based MCI classification. This empirical ceiling aligns with the clinical reality that MCI represents a heterogeneous transitional state where structural changes are minimal and overlap with normal aging. Rather than pursuing diminishing returns through architectural complexity, our hybrid approach explicitly acknowledges this ceiling and builds a clinical fail-safe around it. The MMSE gate (Table IV) triggers alerts on 34% of test cases (29/86 with available MMSE), with 17 REVIEW and 12 ESCALATE alerts that surface imaging–cognition conflicts for clinician attention without overriding the neural model.

B. Accuracy-to-Size Ratio

LightAlzNet's performance is notable: 49.81% F1 with 0.63M parameters. GhostNetV2 (6.18M parameters, 9.8× larger) achieves 48.93% F1—a marginal difference that does not justify its 10× parameter overhead for edge deployment. This validates our thesis that carefully designed lightweight architectures can match or exceed the performance of larger models when purpose-built for the task. Recent work confirms that computationally efficient models (e.g., MobileNet and EfficientNet variants) with explainable AI capabilities represent the most viable path for resource-constrained clinical settings [25].

C. CNN vs. Transformer for Small-Scale Medical Imaging

TinyViT-5M underperforms relative to CNN-based models (46.32% F1, 44.44% accuracy). This suggests that Vision Transformers may require larger datasets than available in typical medical imaging studies to realize their advantage from global self-attention. For moderate-sized cohorts, the locality bias of CNNs remains advantageous.

D. Clinical Safety Through a Symbolic Safety Interlock

The MMSE gate's primary contribution is not in improving raw classification accuracy—it cannot increase the AUC of the visual model—but in adding a layer of clinical safety. By making the system's uncertainty explicit and actionable, the gate ensures that conflicts between imaging and cognitive data are surfaced for human review, the MCI transition zone is handled with appropriate caution, education-related cognitive reserve is accounted for (Rule 6), and clinicians receive structured actionable guidance rather than raw probabilities. The alert-level breakdown (Table IV) provides a quantitative characterization of how often and why the gate intervenes.

E. Gate Limitations and Automation Bias Risk

The gate's recall of 44.2% (19/43 incorrect predictions flagged) means that 55.8% of neural errors pass through undetected. These 24 missed cases represent scenarios where the erroneous imaging prediction happened to be consistent with the MMSE score—for example, a patient incorrectly classified as CN by the neural model whose MMSE also fell in the normal range. This is an inherent limitation of any post-hoc gate that does not have access to independent diagnostic information beyond what the neural model and MMSE already provide. In such cases, neither the imaging model nor the cognitive score independently suggests a problem, so no alert is generated.

The 34% overall trigger rate (29/86 cases) also introduces a risk of alarm fatigue: if one in three routine cases produces a REVIEW or ESCALATE alert, clinicians may begin to discount the gate's output, eroding its safety value. The 65.5% precision—roughly two-thirds of alerts corresponding to actual errors—mitigates this risk but does not eliminate it.

A further concern is automation bias: the presence of a "safety gate" may increase clinician trust in the system's output, including its false negatives. If a clinician receives a NONE alert and therefore does not scrutinize the neural prediction, the 55.8% of errors that pass through the gate undetected become more dangerous than if no gate were present. Any clinical deployment must therefore present the gate's confidence in its own assessment alongside each alert, and the system should never replace clinical judgment.

F. Accuracy in Context: The Data Leakage Ceiling

The 55.74% ensemble accuracy and 0.6886 AUC must be evaluated against the known landscape of ADNI-based classification. Samper-González et al. [22] established the initial reproducibility framework for conventional machine learning on ADNI, which Wen et al. [21] later extended to deep learning in a landmark study demonstrating that many papers reporting 90%+ accuracy on 3-class ADNI tasks rely on data leakage—randomly splitting slices rather than patients, allowing correlated slices from the same scan to appear in both training and test sets. When strict patient-level stratification is applied, reported accuracies converge to the 55–65% range that we observe. Our 5-fold patient-stratified cross-validation ensures that no inter-slice correlations inflate our metrics. The 49.81% F1 of LightAlzNet and 48.68% ensemble F1—well above the 33% random baseline—represent a mathematically significant extraction of biological signal. A 48% F1 on leakage-controlled 3-class medical imaging confirms that meaningful structural patterns are present in the MRI data; the remaining uncertainty is a biological, not an algorithmic, limitation.

G. Limitations

Several limitations should be acknowledged. First, the test set (n=88) is modest, drawn from a single dataset (ADNI), and no external validation on an independent cohort (e.g., OASIS or AIBL) was performed. The reported accuracies may not generalize to other acquisition protocols, populations, or scanner manufacturers.

Second, the ensemble weights were derived from test-set F1 scores rather than a held-out validation set. This introduces optimistic bias in the ensemble's reported metrics. A replication using uniform weights on the 86-image gate-evaluation subset yielded 42.83% macro-F1 versus 49.68% for the weighted variant, confirming that weight selection substantially affects ensemble performance. A production system would determine weights from validation data.

Third, the symbolic gate's recall of 44.2% means most neural errors pass through undetected. The gate is a safety interlock, not a diagnostic system, and its NONE output should never be interpreted as confirmation of correctness. The automation bias risk discussed above further underscores that the system is a decision support tool, not a replacement for clinical judgment.

Fourth, the MMSE is the sole cognitive assessment in the gate. MMSE has well-documented ceiling effects for MCI detection, is confounded by age, education, language, culture, depression, and medications, and has been largely supplanted by the Montreal Cognitive Assessment (MoCA) in specialty memory clinics. The gate's exclusive reliance on MMSE limits its clinical validity, particularly for MCI-range patients. Age-stratified norms and a confounder checklist should be incorporated before deployment.

Fifth, the education-based override (Rule 6) uses a binary ≥ 16 year threshold that oversimplifies cognitive reserve. Education quality, literacy, and cross-national differences in educational systems are not captured.

Sixth, the system uses 2.5D rather than true 3D analysis, which may miss volumetric patterns detectable only through full-volume processing. No 3D baseline was included to quantify this trade-off. Additionally, the 2.5D slice positions (35–65% axial, 45–55% coronal) were chosen based on anatomical priors rather than validated through ablation, so the optimality of this configuration is not established.

Seventh, inference latency (10 ms) was measured on a single desktop CPU; performance on lower-end edge hardware (ARM Cortex, Intel Atom) may be substantially slower.

Finally, no formal significance tests were performed between architectures. Reported F1 differences (e.g., LightAlzNet 49.81% vs. GhostNetV2 48.93%) fall within overlapping 95% confidence intervals and are not statistically significant.

VI. Conclusion

This paper demonstrates that lightweight, 2.5D architectures can extract diagnostically meaningful biological signals from complex neuroimaging data without requiring cloud-scale GPUs. Our custom LightAlzNet achieves the best single-model F1 (49.81%) at a fraction of the parameter count of competing architectures, compressing to 2.03 MB with 10 ms CPU inference latency.

More importantly, we show that combining these edge-optimized models with a symbolic heuristic gate—encoding established clinical knowledge as deterministic rules—yields a more transparent diagnostic tool. By cross-referencing neural predictions against MMSE thresholds across seven clinical rules, the system flags imaging-cognition conflicts, handles the MCI trap zone with appropriate caution, and accounts for cognitive reserve. We also transparently quantify the gate's limitations: a 44.2% recall for detecting neural errors and a 34% alert rate that introduces automation bias risk—trade-offs inherent to any post-hoc safety interlock that must be managed in clinical deployment.

The MCI classification ceiling on structural MRI (∼0.69 AUC) is not a limitation to be overcome through architectural complexity, but a biological reality to be managed through system architecture. A decoupled hybrid design—combining neural perception with a symbolic rule-based safety interlock—offers a path toward AI systems that are both powerful and clinically safe.

Future work will extend the heuristic gate to additional modalities (APOE4 genotype, CSF biomarkers), explore large language model integration for clinical report generation, and validate the system in prospective clinical studies.

Code and Model Availability

The source code, trained model checkpoints, and preprocessing pipeline are available at https://github.com/bmo1177/NeuroDetectLite under the MIT license. All experiments use publicly available ADNI data (https://adni.loni.usc.edu).

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